Ayush Khati

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Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, C++
Backend: FastAPI, Flask, Django, DRF, SQLAlchemy, Celery, REST APIs
Data/Infra: PostgreSQL, MySQL, PostGIS, Redis, Elasticsearch, Azure Event Hubs, Docker, AWS, Linux, Nginx, GitHub Actions
Engineering: Distributed systems, event-driven architecture, concurrency, query optimization, observability, schema migrations, pytest, CI/CD

Experience

Udeck Services – *Software Engineer, Backend* Siliguri, India | Oct 2025 – Present

- Own backend of **Evan Merchantile** (evanmerchantile.in), a live direct-selling commerce platform (**198 endpoints**); kept business logic in a shared service layer used by API, Celery workers and CLI.
- Cut an admin endpoint from **133 queries / 3.96 s to 6 queries / 1.15 s** (**95.5% fewer queries, 71% lower latency**) by modelling a 20-level distributor tree as a **closure table** with indexed reads.
- Made distributor payouts **safe to retry** (**idempotency keys**, unique constraints, **double-entry ledger**, **row-level locks**); fixed concurrent-write race, verified with **500 concurrent requests, zero ledger drift**.
- Designed an event-driven order pipeline on **Celery + Redis** with **idempotent consumers**, exponential backoff and a **dead-letter queue**: **1.2M events/day** with zero lost or duplicated messages during worker crashes.
- Cut **p99 latency from 1.8 s to 240 ms** on a hot read path (**2,500+ accounts, 10,000+ SKUs**) via query-plan profiling, composite indexes and a **cache-aside Redis layer**.
- Led a **zero-downtime migration** of a **38M-row table** (expand/contract, batched backfills, rollback plan); configured **point-in-time recovery (PITR)** for primary database.
- Built **observability** from scratch (logs, metrics, p95/p99 alerts), cutting **MTTD from 45 to 3 minutes**.
- Implemented audit middleware to catch unauthorized credential changes; secured authentication using **Argon2** and rotating **JWT refresh tokens**.

Udeck Services – *Software Engineer Intern, Backend* Siliguri, India | Apr 2025 – Oct 2025

- Built B2B wholesale checkout for **HamroBazaar** (hamrobazaar.in) with tiered pricing, GST slabs, and a double-entry store-credit ledger.
- Cut homepage database queries from **368 to single digits** via bulk preloading and SQLAlchemy eager loading.

Projects

MatchLens: Event-Driven Football Intelligence Platform Oct 2026
Microsoft Inside the Game Hackathon | Main Demo | Streaming Experience | *Python, FastAPI, Pydantic, Next.js, Microsoft Agent Framework, Azure Event Hubs, Microsoft Fabric*

- Built full-stack platform turning typed match events into **deterministic tactical states and evidence-linked explanations** across 4 views; agentic AI output validated application-side.
- Made **Azure Event Hubs** pipeline duplicate-safe (versioned envelopes, sequence tracking, run-scoped reconciliation); verified **1,728/1,728 events processed, zero errors, 1,728 unique IDs** in Microsoft Fabric Eventhouse.

Sentinel: Real-Time CCTV Video Platform Sep 2026
Gujarat Police CCTV-integration challenge | *Python, FastAPI, PostGIS, PyAV, ONNX, React, Docker*

- Built **15-camera** live ingestion service (**RTSP/TCP, HLS**, PyAV decoding) where one failing feed cannot stall the rest: per-camera workers, **2–30 s exponential backoff**, and bounded queues dropping stale frames.
- Added **ONNX Runtime** detection with tracking, **PostGIS** geolocation and live health over SSE.

VibeSec: Static Security Scanner for AI-Generated Code May 2026
Open source: PyPI Package | GitHub Repository | *Python, AST Parsing, Click, pytest*

- Published PyPI CLI (**1,200+ downloads**) catching **24 vulnerability patterns** in AI code (leaked secrets, disabled RLS, XSS) with deterministic **AST engine** and **179 tests**.

Education

Siliguri Institute of Technology – Bachelor of Computer Applications (BCA), CGPA 8.1/10 Expected Jul 2027

Open Source and Leadership

- **Fedora Project contributor** (Mar–May 2025): bug fixes, features and documentation updates, all merged upstream.
- **Technical Lead, TechFrenzy / NSS** (2023–2025): organized **5+ hackathons** for **200+ participants**.